# C S Vaishakh

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## Professional Summary

Artificial Intelligence & Data Science undergraduate with an interest in full-stack development and machine learning. Experienced in building applications using TypeScript (React, Next.js, Express.js etc.), backend systems with Go (Fiber and Gin, Postgres), AI/ML applications in Python using FastAPI and related machine learning frameworks.

## Education

**St. Joseph's College of Engineering and Technology, Palai** Kottayam, Kerala
*B.Tech in Artificial Intelligence and Data Science — APJ Abdul Kalam Technological University* *2023 – 2027*

## Experience

**Intern** — The Nexus Project, IEDC, St. Joseph's College of Engineering and Technology, Palai June 2025
- Built a responsive landing page for HackSphere 2025, a fictional hackathon, in React, TypeScript and Vite.

## Projects

**Quickhand** | *Go, Fiber v3, WebSockets, TanStack Start, PostgreSQL, Docker* December 2025 – Present
- Implemented role-based access control across two user roles, securing Fiber v3 routes with stateless JWTs.
- Built an authenticated WebSocket layer broadcasting real-time job alerts to matched handymen.
- Structured the codebase as a Go workspace monorepo pairing the Fiber API with a TanStack Start frontend.

**Snip** | *Go, Gin, Redis, PostgreSQL, React, AWS EC2, Vercel* July – August 2026
- Built a URL shortener with a Go (Gin) REST API, React dashboard, JWT auth, and collision-safe codes.
- Cut redirect latency by fronting PostgreSQL with a Redis cache on the public redirect path.
- Deployed the frontend on Vercel and the backend on AWS EC2 behind Caddy via GitHub Actions CI/CD.

**PortfolyoIQ** | *Next.js, Express.js, FastAPI, TensorFlow.js, PostgreSQL, AWS EC2* March – August 2026
- Engineered a privacy-preserving rebalancing platform: Next.js client, Express API, FastAPI model service.
- Trained logistic regression models in-browser with TensorFlow.js, eliminating raw portfolio data uploads.
- Implemented Federated Averaging (FedAvg) on the Express aggregator to synchronize local model updates.
- Self-hosted all three services on AWS EC2 behind Caddy via systemd, deployed by GitHub Actions.

**ONT Monitor** | *Go, Fiber v3, GORM, PostgreSQL, Twilio API, Docker* April 2026
- Built a Go microservice (Fiber v3, GORM) ingesting real-time ONT status change events.
- Implemented thread-safe in-memory queues and a worker pool to evaluate outages without race conditions.
- Integrated the Twilio API to dispatch outage and recovery WhatsApp alerts on state transitions.

## Technical Skills

**Programming**: Python, TypeScript, Go, Java, JavaScript
**Web & Backend**: Next.js, React, Express.js, Fiber v3, Gin, FastAPI, Node.js, REST APIs, WebSockets
**Databases & ORMs**: PostgreSQL, MongoDB, Redis, GORM, Drizzle ORM, Mongoose
**AI / ML**: TensorFlow.js, scikit-learn, NumPy, Pandas, Feature Engineering, Model Evaluation
**Tools & DevOps**: Git, GitHub Actions (CI/CD), Docker, AWS EC2, Caddy, Vercel, Bun, Linux, Postman

## Achievements & Leadership

**ASTHRA 11.0 Student Coordinator**: one of 4 leading the fest; set up 6 teams, co-managed leads (Sep 2026)
**Third Place, Portfolyo**: portfolio website challenge, IEEE Student Branch SJCET Palai (June 2025)
**NPTEL Certification**: Data Science Using Python, 12-week course (July – October 2024)